Neel Gali

Dexterity Engineer

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PROFESSIONAL SUMMARY

Dexterity Engineer with 2+ years of professional experience developing real-time, hardware-in-the-loop software for complex robotic systems, complemented by expertise in AI/ML. Proven ability to build robust teleop stacks, integrate haptic frameworks, and design simulation and testing infrastructure using Python and cloud-native platforms. Skilled in developing training-ready data pipelines for autonomy and reinforcement learning pipelines to achieve bimanual manipulation.

TECHNICAL SKILLS

Programming & Scripting: Python, C++, Bash, SQL

Robotics & Control: ROS, Real-time Systems, Hardware-in-the-Loop Software Development, Controls for Robotics Applications, Teleoperation, Dexterous Manipulation, Bimanual Manipulation, Kinematics

Simulation & Haptics: MuJoCo, IsaacSim, Haptic Frameworks, Tactile Sensor Data Integration, Force-Reflective Feedback, Automated Workflows, VR Peripheral Integration (Unity, OpenXR)

Cloud & DevOps: Docker, Containerization, Kubernetes, AWS, Azure, Git, CI/CD

AI & Machine Learning: Reinforcement Learning Pipelines, Autonomy, Training-ready Data, Learned Policies, Computer Vision (OpenCV), Machine Learning

PROFESSIONAL EXPERIENCE

AI / ML Engineer | Epik Solutions

Jun 2025 - Present

• Engineered robust software pipelines in Python to process high-volume sensor and telemetry data, ensuring real-time signal integrity for critical cybersecurity applications, reducing false positives by 15% for security teams.

• Collaborated with cross-functional teams to translate complex human intent into specific robotic execution requirements, defining robust control methodologies and evaluation metrics for hardware-in-the-loop software development.

• Developed advanced computer vision models using PyTorch and OpenCV to enable precise object detection and tracking, generating training-ready data for advanced autonomy and reinforcement learning pipelines in dynamic environments.

• Designed and implemented data ingestion and transformation workflows for high-bandwidth tactile sensor data, ensuring clean, structured inputs for both model training and real-time control loops.

• Implemented robust algorithms for anomaly detection and jitter reduction within teleoperation data streams, enhancing the transparency and reliability of robotic execution in complex, contact-rich tasks.

• Developed Bash scripts and automated workflows for deploying and monitoring real-time robotic control systems, streamlining production environments and reducing deployment time by 20%.

AI / ML Engineer | Samay Systems LLC

July 2024 - Jun 2025

• Engineered and refined retargeting algorithms (IK-based and optimization-based) for compliance robotics, effectively mapping human hand kinematics to varied robotic configurations for high-fidelity execution.

• Developed Python services to collect and process multi-modal sensor data from AWS and Azure cloud platforms, ensuring consistent inputs for training-ready data used in shared autonomy initiatives.

• Integrated advanced haptic frameworks and tactile sensor data into existing control loops, providing operators with critical force-reflective feedback during intricate dexterous manipulation tasks.

• Designed and implemented a robust simulation and testing infrastructure using Docker containers, validating end-effectors and evaluating performance metrics for robotic control systems in various compliance scenarios.

• Optimized SQL workflows to manage and log high-bandwidth tactile data, creating scalable data management solutions crucial for future reinforcement learning pipelines.

• Spearheaded the deployment of real-time robotic software services utilizing AWS cloud-native infrastructure, Docker, and Kubernetes, ensuring reliable operation of contact-rich tasks in production environments.

ML Intern | Monarch Tractor

Jun 2023 - Aug 2023

• Supported the core development of robust controls for robotics applications within semi-autonomous agricultural vehicles, focusing on enhancing the teleop stack for improved operator transparency.

• Developed Python-based preprocessing pipelines to transform raw sensor and image data into training-ready data, specifically for bimanual manipulation and object detection tasks in complex field environments.

• Implemented object detection and segmentation models using PyTorch and OpenCV, validating model outputs against real-world operational constraints to improve robotic execution in autonomous vehicle perception systems.

• Conducted extensive real-robot testbed validation for end-effectors using CLI-driven workflows, contributing to the performance metrics and dexterity scoring of the robotic system.

• Collaborated cross-functionally to refine requirements for retargeting algorithms and haptic integration, aligning robotic capabilities with human intent for field deployment.

• Deployed and monitored robotic control software in cloud-hosted production environments, gaining practical experience with real-time performance monitoring and validation of learned policies.

Associate Data Scientist | ZeMoSo Technologies

Jul 2021 - Aug 2022

• Contributed to a cloud-based platform, developing scalable systems in Python to manage experiment and sample data, forming a foundational understanding for reliable data pipelines.

• Developed backend services and automated workflows for data validation and structuring, ensuring clean inputs essential for training-ready data for future autonomy and reinforcement learning pipelines.

• Supported proof-of-concept for real-time control experiments by automating data preparation and feature generation, directly contributing to early-stage development of robotic execution strategies.

• Integrated CI/CD pipelines for software deployment and version control, enforcing quality checks and maintaining auditability crucial for reliable and scalable systems in regulated environments.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India